Hugo Marquerie 12/01/2026

Lem riemann lebesgue l1

Lema 1 (de Riemann-Lebesgue). *Sea* $f \in \mathcal{L}^1([0,1))$, *entonces*

$$\lim_{|n|\to\infty} \left|\widehat{f}(n)\right| = 0.$$

***Demostración*:** Partiendo de la identidad de Euler, tenemos que

$$\begin{aligned}
\widehat{f}(n) &= \int_0^1 f(x)e^{-2\pi inx}\,\mathrm{d}x = -\int_0^1 f(x)e^{-2\pi inx}e^{-\pi i}\,\mathrm{d}x = -\int_0^1 f(x)e^{-2\pi in(x+1/2n)}\,\mathrm{d}x \\
&= -\int_{\frac{1}{2n}}^{1+\frac{1}{2n}} f\left(y - \frac{1}{2n}\right)e^{-2\pi iny}\,\mathrm{d}y = -\int_0^1 f\left(x - \frac{1}{2n}\right)e^{-2\pi inx}\,\mathrm{d}x. \\
\implies \widehat{f}(n) &= \frac{1}{2}\left[\int_0^1 f(x)e^{-2\pi inx}\,\mathrm{d}x - \int_0^1 f\left(x - \frac{1}{2n}\right)e^{-2\pi inx}\,\mathrm{d}x\right] \\
&= \frac{1}{2}\int_0^1 \left[f(x) - T_{1/2n}f(x)\right]e^{-2\pi inx}\,\mathrm{d}x
\end{aligned}$$

Por el `lem-convergencia-lp-traslacion`/Lema 1, tenemos que

$$0 \leqslant \lim_{|n|\to\infty}\left|\widehat{f}(n)\right| \leqslant \lim_{|n|\to\infty}\frac{1}{2}\int_0^1\left|f(x) - T_{1/2n}f(x)\right|\mathrm{d}x = 0.$$

■

Referenciado en

- `prop-criterio-dini`
- `lem-serie-fourier-derivada`

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.